

## HERMANUS, South Africa

WMO: 689180

Lat: 34.4325S

Lon: 19.2247E

Elev: 16

StdP: 101.13

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 8.5        | 9.4 | 3.0                           | 4.7 | 14.7 | 4.4 | 5.2 | 12.8 | 11.4                  | 15.9 | 10.4 | 15.0 | 1.8                   | 70   | 0.401 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 4.8                    | 26.2            | 18.9 | 24.8 | 18.8 | 23.7 | 18.7 | 20.9                | 23.6 | 20.3 | 23.0 | 19.7 | 22.4 | 4.1                  | 110  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 20.0                            | 14.7 | 22.3 | 19.3 | 14.1 | 21.7 | 18.7 | 13.6 | 21.3 | 60.4          | 23.7 | 58.5 | 23.2 | 56.5 | 22.5 | 24.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 10.5              | 9.3  | 8.4 | 7.2                        | 29.8 | 0.6                | 1.5 | 6.8                                                | 30.8 | 6.4        | 31.7 | 6.1        | 32.5 | 5.6        | 33.5 |    |    |
|                   |      |     | 6.0                        | 21.5 | 0.8                | 0.8 | 5.4                                                | 22.1 | 4.9        | 22.6 | 4.4        | 23.0 | 3.8        | 23.6 |    |    |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 17.2   | 20.2  | 20.1  | 19.2  | 17.8  | 16.2  | 14.7  | 14.1  | 14.3  | 15.3  | 16.9  | 18.1  | 19.7 |
|                                                                   | DBStd       | 2.86   | 1.58  | 1.45  | 1.94  | 2.00  | 1.89  | 1.84  | 2.06  | 1.96  | 1.92  | 1.98  | 1.70  | 1.54 |
|                                                                   | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                   | HDD18.3     | 663    | 2     | 2     | 12    | 33    | 70    | 111   | 132   | 127   | 94    | 53    | 23    | 4    |
|                                                                   | CDD10.0     | 2629   | 317   | 283   | 286   | 233   | 193   | 141   | 128   | 133   | 158   | 214   | 243   | 300  |
|                                                                   | CDD18.3     | 251    | 61    | 52    | 39    | 16    | 4     | 2     | 2     | 1     | 2     | 9     | 16    | 46   |
|                                                                   | CDH23.3     | 334    | 73    | 43    | 55    | 35    | 8     | 8     | 16    | 6     | 6     | 21    | 19    | 44   |
| Wind                                                              | WSAvg       | 3.6    | 3.8   | 3.5   | 3.5   | 3.1   | 3.1   | 3.4   | 3.4   | 3.7   | 3.8   | 4.1   | 4.1   | 3.8  |
|                                                                   | PrecAvg     | 633    | 24    | 22    | 26    | 56    | 70    | 95    | 92    | 78    | 50    | 48    | 43    | 28   |
|                                                                   | PrecMax     | 943    | 104   | 68    | 102   | 179   | 146   | 205   | 168   | 240   | 125   | 141   | 129   | 94   |
|                                                                   | PrecMin     | 343    | 6     | 1     | 6     | 12    | 10    | 19    | 12    | 24    | 7     | 4     | 3     | 5    |
|                                                                   | PrecStd     | 116    | 23    | 15    | 20    | 41    | 29    | 42    | 42    | 43    | 27    | 34    | 37    | 21   |
|                                                                   | DB          | 27.2   | 26.6  | 27.4  | 26.7  | 24.7  | 24.1  | 25.1  | 24.3  | 23.6  | 26.3  | 25.7  | 26.5  |      |
|                                                                   | MCWB        | 20.1   | 19.7  | 19.6  | 17.6  | 15.4  | 14.0  | 13.8  | 13.5  | 16.2  | 18.8  | 18.1  | 19.7  |      |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 2% DB       | 25.0   | 24.6  | 24.7  | 23.7  | 21.5  | 20.2  | 21.1  | 19.9  | 20.8  | 22.9  | 23.3  | 24.6  |      |
|                                                                   | 2% MCWB     | 19.9   | 19.4  | 19.0  | 17.6  | 15.8  | 13.8  | 13.2  | 14.0  | 15.5  | 17.6  | 17.6  | 19.1  |      |
|                                                                   | 5% DB       | 23.8   | 23.5  | 23.1  | 21.7  | 20.2  | 18.5  | 18.5  | 18.3  | 19.2  | 21.2  | 21.9  | 23.3  |      |
|                                                                   | 5% MCWB     | 19.4   | 19.2  | 18.7  | 17.2  | 15.5  | 14.0  | 13.7  | 13.8  | 15.0  | 16.6  | 17.5  | 18.7  |      |
|                                                                   | 10% DB      | 22.9   | 22.7  | 22.0  | 20.5  | 19.2  | 17.5  | 17.2  | 17.1  | 18.0  | 19.9  | 20.9  | 22.4  |      |
|                                                                   | 10% MCWB    | 18.9   | 18.9  | 18.4  | 16.9  | 15.3  | 13.8  | 13.5  | 13.4  | 14.4  | 16.0  | 17.0  | 18.2  |      |
|                                                                   | WB          | 21.7   | 21.6  | 21.1  | 19.9  | 17.6  | 16.4  | 16.2  | 16.5  | 17.6  | 19.7  | 20.3  | 21.0  |      |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | MCDB        | 24.4   | 23.6  | 24.2  | 22.7  | 20.1  | 18.7  | 20.1  | 19.3  | 20.3  | 23.7  | 22.7  | 24.3  |      |
|                                                                   | 2% WB       | 20.9   | 20.7  | 20.3  | 18.9  | 17.0  | 15.6  | 15.1  | 15.6  | 16.5  | 18.4  | 19.1  | 20.0  |      |
|                                                                   | 2% MCDB     | 23.8   | 22.8  | 23.2  | 21.4  | 19.1  | 17.7  | 18.0  | 17.8  | 19.3  | 22.0  | 21.5  | 23.3  |      |
|                                                                   | 5% WB       | 20.2   | 20.1  | 19.6  | 18.2  | 16.5  | 15.0  | 14.4  | 14.6  | 15.7  | 17.4  | 18.4  | 19.3  |      |
|                                                                   | 5% MCDB     | 22.9   | 22.4  | 22.3  | 20.6  | 18.7  | 17.2  | 17.2  | 17.3  | 18.4  | 20.4  | 20.8  | 22.4  |      |
|                                                                   | 10% WB      | 19.5   | 19.5  | 18.9  | 17.5  | 16.0  | 14.4  | 13.7  | 13.8  | 14.9  | 16.6  | 17.6  | 18.7  |      |
|                                                                   | 10% MCDB    | 22.2   | 21.9  | 21.5  | 19.9  | 18.3  | 16.7  | 16.5  | 16.5  | 17.6  | 19.3  | 20.2  | 21.7  |      |
| Mean Daily Temperature Range                                      | MDBR        | 4.8    | 4.7   | 4.9   | 5.4   | 5.3   | 5.7   | 5.8   | 5.6   | 5.3   | 5.2   | 5.0   | 4.9   |      |
|                                                                   | 5% DB       | 6.4    | 6.1   | 6.5   | 7.3   | 6.6   | 7.2   | 8.0   | 7.3   | 6.9   | 7.5   | 6.8   | 6.3   |      |
|                                                                   | MCWBR       | 3.0    | 2.9   | 3.0   | 3.3   | 3.4   | 3.8   | 4.0   | 3.7   | 3.7   | 3.7   | 3.4   | 3.3   |      |
|                                                                   | 5% WB       | 5.3    | 4.6   | 5.7   | 6.1   | 5.4   | 5.5   | 6.8   | 6.2   | 6.4   | 6.6   | 5.4   | 5.6   |      |
|                                                                   | MCWBR       | 2.9    | 3.0   | 3.1   | 3.3   | 3.5   | 3.6   | 4.2   | 3.7   | 3.8   | 3.7   | 3.4   | 3.2   |      |
| Clear-Sky Solar Irradiance                                        | taub        | 0.352  | 0.354 | 0.344 | 0.328 | 0.323 | 0.316 | 0.318 | 0.335 | 0.347 | 0.360 | 0.347 | 0.352 |      |
|                                                                   | taud        | 2.523  | 2.527 | 2.551 | 2.587 | 2.585 | 2.578 | 2.565 | 2.510 | 2.453 | 2.434 | 2.510 | 2.514 |      |
|                                                                   | Ebn at Noon | 984    | 961   | 935   | 895   | 843   | 822   | 839   | 872   | 915   | 944   | 983   | 988   |      |
|                                                                   | Edh at Noon | 111    | 107   | 97    | 83    | 74    | 70    | 74    | 88    | 105   | 116   | 112   | 113   |      |
| All-Sky Solar Radiation                                           | RadAvg      | 7.79   | 6.92  | 5.56  | 4.08  | 2.95  | 2.48  | 2.74  | 3.51  | 4.82  | 6.19  | 7.36  | 7.92  |      |
|                                                                   | RadStd      | 0.23   | 0.24  | 0.28  | 0.19  | 0.16  | 0.14  | 0.13  | 0.17  | 0.22  | 0.32  | 0.27  | 0.30  |      |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (12 neighbors) | +0.36 | +0.49   | N/A    | N/A     | +0.53 | +0.47 | -2          | -74     | +111    | +61     |

Nomenclature: See separate page